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TEF-RAG: Temporal Evidence Flow Retrieval-Augmented Generation for Energy Storage Station Operation and Maintenance

Juxian Yin

***Abstract*—For embodied operation and maintenance (O&M) in energy-storage stations, a diagnosis or action plan often depends on alarms, inspections, work orders, procedures, and post-action checks. These records change over time. Retrieval-augmented generation (RAG) must therefore retrieve records available at the query time and enough complementary evidence to support diagnosis, action, and verification. Using later records leaks retrospective information; omitting a process stage leaves the decision context incomplete. We propose TEF-RAG (Temporal Evidence Flow Retrieval-Augmented Generation) to select query-conditioned temporal evidence flows. Deterministic checks enforce asset identity, event time, availability time, and procedure applicability. A learned pair proposer selects evidence pairs for semantic relation judging. The accepted relations, including support, update, qualification, supersession, and verification, form a directed graph. A nonlinear RankNet then compares candidate evidence sets and selects at most five records for complementary roles and coherent relations. We evaluate the method on a public-source-grounded, AI-assisted synthetic temporal O&M benchmark containing dual timestamps, supersession, cross-source complementary evidence, and persistent uncertainty. Six retrieval methods are compared with a shared downstream generator. TEF-RAG has the highest Recall@5, Complete@5, and FlowComplete@5, while Fine-tuned BGE has the highest nDCG@5. TEF-RAG also has the highest Field Macro F1, Action F1, Citation F1, and Evidence Support Recall for structured generation. The comparison separates the quality of individual-record ranking from the recovery of temporally valid, process-complete evidence sets.**

***Index Terms*—retrieval-augmented generation, temporal retrieval, evidence flow, learning to rank, operation and maintenance, energy storage**

I. Introduction

Handling an anomaly in an energy-storage station requires records from several stages of operation and maintenance (O&M): equipment states, alarms, inspections, earlier work orders, procedures, and post-action checks. An embodied O&M system must select from these records using the knowledge available when the query was issued. It must also recover the evidence needed to connect the observed state to a diagnosis, an action, and verification of that action. Retrieval gives the generator access to these changing records. Retrieval-augmented generation (RAG) combines language models with external retrieval for knowledge-intensive generation [**?**]. Best Matching 25 (BM25) is a strong lexical baseline; DPR and ColBERT use dense representations and late interaction, respectively, for semantic matching [**?**], [**?**]. BEIR compares how retrieval methods generalize across domains [**?**]. At the generation stage, FiD aggregates multiple retrieved passages [**?**], and Self-RAG jointly controls retrieval, generation, and self-reflection [**?**]. Conventional RAG usually ranks candidate documents independently. Two properties of O&M records complicate this approach. A record has both an event time and an availability time: an event may have occurred before a query even though its record entered the system afterward. Such a record is unavailable to the earlier query. Maintenance decisions also require complementary evidence. An observation motivates an inspection, which supports a diagnosis and an action; a later check verifies the outcome. A Top-k set of individually relevant records can still miss part of this process.

Time-aware RAG handles temporal constraints and changes in information over time [**?**], [**?**]. Multi-hop methods condition retrieval on evidence found in earlier steps [**?**], [**?**], while RAPTOR, GraphRAG, and HippoRAG organize documents through hierarchies or graphs [**?**], [**?**], [**?**]. These methods address parts of the O&M retrieval problem, but the cited work does not explicitly combine query-time visibility with set-level optimization for process completeness. We propose TEF-RAG (Temporal Evidence Flow Retrieval-Augmented Generation) to select evidence sets under both requirements. Deterministic asset, temporal, and procedure checks first remove unusable records. A learned proposer selects evidence pairs for query-conditioned relation judging, which produces a directed temporal evidence graph. A nonlinear pairwise ranker then compares candidate evidence sets using the annotated completeness objective.

TEF-RAG constructs its graph from the records visible to the current query. The edges describe task-specific relations among records, including branching, updates, supersession, and unresolved uncertainty. Temporal constraints determine which records are eligible; the final ranker selects among eligible sets for complementary roles and consistent relations. This separates the visibility check from the set-selection objective and allows evidence structures beyond a single retrieval path. The selected records provide context for a shared generator that produces a standardized work order and a dependency-aware action plan. These outputs are intended as an interface to downstream planners, constraint checkers, and robotic execution. We evaluate evidence retrieval and structured planning context; closed-loop control and field safety certification fall outside this evaluation.

The paper makes three contributions:

1) We formulate temporal evidence flow retrieval for dynamic O&M records, with explicit requirements for

query-time visibility and evidence-set completeness.
2) We combine learned evidence-pair proposal, query-conditioned relation graphs, and nonlinear set ranking to select records that jointly support the required process.
3) We construct a controlled temporal O&M benchmark and compare retrieval and evidence-grounded structured generation with a shared downstream generator.

II. Related Work

Retrieval-Augmented Generation and Neural Retrieval. RAG couples an external retriever with a generator [**?**]. DPR uses dense dual encoders, and ColBERT uses late interaction to rank passages [**?**], [**?**]. FiD aggregates retrieved passages at generation time [**?**]; BEIR compares sparse, dense, and reranking methods across domains [**?**]. TEF-RAG assigns retrieval utility to a set of records together with their temporal and semantic relations.

Time-Aware and Multi-Hop Retrieval. Time-aware retrieval selects evidence under a temporal condition, while multi-hop retrieval seeks records whose usefulness depends on other evidence. MRAG and TA-RAG incorporate temporal conditions into retrieval [**?**], [**?**]. MDR retrieves evidence sequentially, and IRCoT interleaves retrieval with reasoning [**?**], [**?**]. TEF-RAG requires both temporal validity and complementary records. It distinguishes event time from availability time and represents their task-specific relations in a directed graph that can contain more than a linear chain.

Structured and Graph-Based RAG. RAPTOR, GraphRAG, and HippoRAG organize information across documents through hierarchies or graphs [**?**], [**?**], [**?**]. TEF-RAG builds edges from the current query and the visible O&M records. Its edges encode process relations among evidence records, whereas document hierarchies and entity relations provide the organizing structures in much of this work.

Embodied O&M and Battery Intelligence. Retrieved operational knowledge also supports field robotics and equipment management. SayComply retrieves compliance knowledge for field robots, and RAGLRO applies retrieval-augmented language models to robotic operation tasks [**?**], [**?**]. LiBrain combines time-series information, retrieved knowledge, and language models for battery diagnosis and management [**?**]. TEF-RAG addresses the evidence-selection step before such downstream tasks: the selected records must be visible at the query time and cover the required process roles.

Selecting a final evidence set also requires a way to compare alternative combinations of records. **Learning to Rank.** TEF-RAG treats final set selection as a query-level learning-to-rank problem. RankNet learns ranking functions from pairwise preferences [**?**], and ANCE uses hard negatives to learn retrieval boundaries [**?**]. We train on pairs of evidence sets for the same query, comparing complete sets with high-scoring alternatives that lack required evidence or relations.

III. Problem Definition

A. Queries and Temporally Visible Evidence

We represent an O&M query as

$$Q_q = (x_q, u_q, m_q, t_q), \tag{1}$$

where x_q is the query text, u_q is the target asset, m_q denotes the asset-model/context metadata used for scope checks, and t_q is the knowledge cutoff time. The global evidence repository is denoted by $\mathcal{K} = \{e_1, \ldots, e_N\}$. Each evidence record contains at least its content, asset identifier, event type, source type, event time $t_e(e)$, and availability time $t_a(e)$.

We explicitly distinguish between when an event happened and when the system knew about it. Evidence e is visible to query q if and only if

$$\mathrm{Visible}(e, q) = \mathbb{I}[t_e(e) \le t_q \wedge t_a(e) \le t_q]\,. \tag{2}$$

Thus, a record describing an earlier event but entered into the system only after the query time cannot be used by that query.

For procedure evidence, we additionally require a matching asset model, validity at the query time, and that the procedure has not been withdrawn. We write these deterministic conditions collectively as

$$\mathrm{Legal}(e, Q_q) = \mathrm{Visible}(e, q) \wedge \mathrm{ScopeOK}(e, Q_q), \tag{3}$$

where ScopeOK summarizes asset consistency and, when applicable, procedure model scope, validity interval, and withdrawal constraints. The legal evidence set at query time is therefore

$$\mathcal{C}_q = \{e \in \mathcal{K} : \mathrm{Legal}(e, Q_q) = 1\}. \tag{4}$$

All subsequent retrieval, relation reasoning, and set ranking operate only within $\mathcal{C}_q$.

To distinguish the levels used below, an *evidence record* refers to one retrievable item, an *evidence set* refers to a selected collection of records, and a *temporal evidence flow* refers to such a set together with the directed relations among its records.

B. Retrieval Objective

A base retriever produces the candidate node pool $V_q \subseteq \mathcal{C}_q$. With a retrieval upper bound of $k = 5$, we set $k_q = \min(5, |V_q|)$ and compare candidate sets containing exactly k_q records. The output therefore contains at most five records. When fewer than five legal records exist, all available legal evidence is returned, without padding or backfilling. Selection favors records that answer the query, cover its required evidence roles, and have consistent temporal and semantic relations.

We select the evidence set by its joint utility:

$$\hat{\mathcal{S}}_q = \arg\max_{\mathcal{S} \in \mathcal{B}_q} F_\theta(Q_q, \mathcal{S}, \mathcal{G}_q), \quad |\mathcal{S}| = k_q, \tag{5}$$

Here, $\mathcal{B}_q$ contains candidate evidence sets of size k_q, $\mathcal{G}_q$ is the query-conditioned temporal evidence flow graph, and F_θ is the learned set-level ranking function.

TABLE I
MAIN NOTATION.

Symbol	Meaning
$Q_q = (x_q, u_q, m_q, t_q)$	Query text, asset, asset-model/context metadata, and knowledge cutoff time
$e, t_e(e), t_a(e)$	Evidence record, event time, and first availability time
$\mathcal{K}$	Global evidence repository
$\mathcal{C}_q$	Legal and visible evidence set at query time
V_q	Candidate node pool returned by the base retrieval stage
$\mathcal{G}_q$	Query-conditioned temporal evidence flow graph
$\mathcal{B}_q$	Candidate space of evidence sets of size $k_q \le 5$
$\hat{\mathcal{S}}_q$	Final selected evidence set (at most 5 records)
W_q, A_q	Standardized work order and action plan

C. Downstream Task

The retrieval result is used as controlled context for a structured generator, which produces a standardized work order W_q and an action plan A_q:

$$(\hat{W}_q, \hat{A}_q) = g_\psi(x_q, \hat{\mathcal{S}}_q). \tag{6}$$

The generator receives the records in $\hat{\mathcal{S}}_q$ as context, without retrieval scores or graph features. The generation experiments test whether evidence selection affects work-order fields, actions, dependencies, and citations. The retrieval method constructs and ranks the temporal evidence flows used for this comparison.

IV. TEF-RAG METHOD

TEF-RAG has five stages (Fig. **??**): legal-candidate retrieval, learned evidence-pair proposal, query-conditioned relation graph construction, candidate evidence-set construction, and nonlinear set ranking. Candidate retrieval limits the node pool, and pair proposal limits the relation-judging budget. The graph records process relations among the retained evidence; candidate-set construction and ranking use these relations to compare alternative sets.

A. Candidate Evidence Preparation and Temporal Legality

TEF-RAG checks each raw record against the target asset, query time, and procedure scope. It excludes records for other assets, future events, records that became available after the query, and procedures outside the applicable model or validity period. These deterministic checks define the legal evidence pool before relevance scoring.

BM25 retrieves a broad pool from $\mathcal{C}_q$. We assign each record an initial node score using lexical relevance, recency, and compatibility with the evidence roles implied by the query. The 30 highest-scoring legal records form V_q (Section **??**). Relation reasoning and final set selection operate on this pool.

Applying semantic relation reasoning to every candidate pair in the search pool would incur quadratic cost in the number of candidates. TEF-RAG therefore first learns a lightweight pair proposer that estimates, for each temporally valid directed pair (e_i, e_j), the probability that the pair is worth passing to the relation-reasoning stage:

$$p_{ij} = \sigma\big(\mathbf{w}^\top \phi_p(q, e_i, e_j) + b\big)\,. \tag{7}$$

The features $\phi_p(q, e_i, e_j)$ describe query relevance for both records, similarity between the records, event types, temporal gaps, and supersession information. A pair is labeled positive if it realizes a required relation in a training reference evidence flow. All other temporally valid pairs are labeled negative. Logistic regression learns the pair priorities, and the highest-scoring pairs proceed to relation judging.

With a Top-30 pool and a single forward temporal direction, there are at most $\binom{30}{2} = 435$ temporally valid pairs. The proposer retains the 32 highest-scoring pairs. For a pool of 30 records, this reduces the number of relation judgments by up to approximately 92.6%. Pair proposal thus allocates the relation-judging budget while retaining the deterministic temporal direction constraint.

B. Query-Conditioned Temporal Evidence Flow

For each retained candidate pair, TEF-RAG uses a query-conditioned large language model (LLM) relation judge that reads the query together with the text and metadata of both evidence records and outputs

$$h_\eta(q, e_i, e_j) \to (r_{ij}, c_{ij}), \tag{8}$$

Here, $r_{ij} \in \mathcal{R}$ is the predicted relation and $c_{ij} \in [0, 1]$ its confidence. We add an edge only when the judge predicts a valid relation above the confidence threshold. Appendix **??** gives the fixed judging prompt.

Candidate evidence records are treated as graph nodes, and identified semantic relations between two records are represented as directed edges. For query q, the temporal evidence flow graph is

$$\mathcal{G}_q = (V_q, E_q), \tag{9}$$

Here, V_q contains the candidate records, and E_q contains the accepted directed relations. An edge (e_i, r, e_j) links source record e_i to target record e_j with relation $r \in \mathcal{R}$.

We use ten process relations: supports, governs, qualifies, contrasts, prerequisite, preserves uncertainty, resolves, supersession, updates, and verifies. Edges follow event time or an explicit prerequisite/update direction. The resulting graph can branch and merge.

The graph $\mathcal{G}_q$ depends on the query: the same records can have different task roles for different queries. Relation judging operates within the legal pool defined in Section **??**, so it cannot reintroduce late-arriving records, future events, or inapplicable procedures.

C. Candidate Evidence-Set Construction

We use the graph to compare combinations of evidence records. Since enumerating all five-record subsets of the search pool is expensive, the final ranker considers only sets generated by the following procedures.

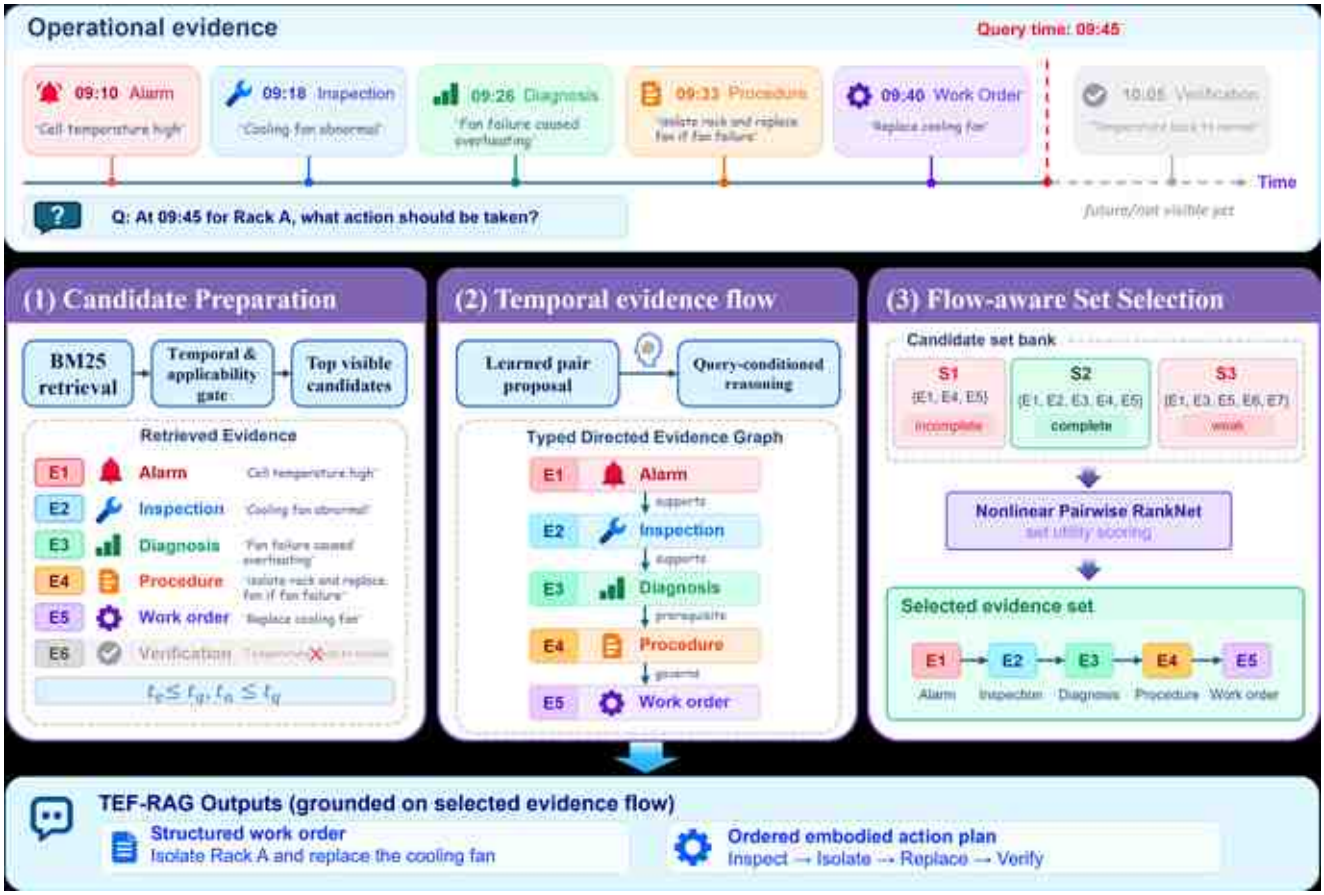


Fig. 1. Overview of TEF-RAG. The system first selects candidate evidence pairs from records that are legal at the query time, then constructs a query-conditioned temporal evidence flow. Candidate evidence sets are constructed from the resulting retrieval and graph signals, and a nonlinear pairwise ranker selects at most five records as the final evidence set for a shared downstream generator.

We enumerate five-record combinations among the highest-scoring nodes. We also retain sets produced by earlier retrieval steps and generate variants by replacing a single record. These replacements allow a lower-ranked record to enter a set when it fills a required task role.

For each set $\mathcal{S}$, we extract a set-level representation $\phi_s(q, \mathcal{S}, \mathcal{G}_q)$. The frozen feature schema used by the final model contains 62 dimensions. The features comprise:

- statistics of node relevance, recency, role compatibility, and aggregate node scores;
- pairwise lexical similarity, event/availability spans, and event-gap statistics;
- event-type and source-type counts and diversity;
- task-role coverage, missing-demand counts, and query/set uncertainty indicators;
- graph structure, including active edges, edge confidence and density, node degrees, connected components, and directed-path statistics;
- the rule-based node, edge, role-coverage, redundancy, connectivity, and uncertainty-consistency terms used during first-round hard-negative selection.

The representation combines record quality with role coverage, internal relations, redundancy, and uncertainty consistency to score the set as a whole.

D. Nonlinear Pairwise Set Ranking

The final ranker is a nonlinear RankNet [**?**]. Let

$$s_\theta(q, \mathcal{S}) = \mathrm{MLP}_\theta(\phi_s(q, \mathcal{S}, \mathcal{G}_q)) . \tag{10}$$

For each query, we pair complete candidate sets with incomplete alternatives. Each training pair is written as $(\mathcal{S}_i^+, \mathcal{S}_j^-)$, where $\mathcal{S}_i^+$ forms a complete reference evidence flow and $\mathcal{S}_j^-$ lacks required evidence or relations.

The pairwise loss favors the complete set:

$$\ell(q, \mathcal{S}_i^+, \mathcal{S}_j^-) = -\log \sigma \left(s_\theta(q, \mathcal{S}_i^+) - s_\theta(q, \mathcal{S}_j^-) \right) . \tag{11}$$

We average this loss over the training pairs:

$$\mathcal{L} = \frac{1}{|\mathcal{P}|} \sum_{(q, \mathcal{S}_i^+, \mathcal{S}_j^-) \in \mathcal{P}} \ell(q, \mathcal{S}_i^+, \mathcal{S}_j^-), \tag{12}$$

where $\mathcal{P}$ denotes all candidate-set pairs constructed during training.

Training has two rounds. The first uses a rule-based set score to prioritize incomplete sets that contain high-quality records, internal relations, and broad role coverage. The score also

accounts for redundancy and isolated records (Appendix **??**). These negatives may look relevant while missing an evidence category or a required relation. We retain a small number of negatives with different compositions to broaden the training examples.

We then rescore all incomplete sets with the first-round ranker. The highest-scoring failures become hard negatives for the second round: they are incomplete sets that the current model would otherwise rank highly. Retraining on these pairs targets errors in distinguishing complete from incomplete evidence flows.

E. Interface to the Embodied O&M Generator

TEF-RAG supplies evidence context to a shared downstream generator. Given the query and selected records, the generator produces a standardized work order and a dependency-aware action plan for use by a planner, constraint checker, or robotic executor.

All methods use the same downstream generator to isolate the effect of evidence selection. It receives the query and selected records, with the method name, retrieval scores, and graph structure withheld. Appendix **??** gives the fixed prompt. The evaluation covers structured work orders and action plans; it excludes closed-loop execution and field safety certification.

V. Experimental Design

A. Temporal O&M Benchmark

We construct a public-source-grounded synthetic temporal O&M benchmark with artificial intelligence (AI) assistance. Public product documents and research datasets constrain cell type, typical physical ranges, and sampling background. We synthesize the O&M event processes, diagnostic changes, procedure contexts, and query tasks under those constraints. This gives us control over late-arriving records, supersession, cross-source complementary evidence, multi-episode ambiguity, and persistent uncertainty. The records are synthetic, not real station fault logs or a real O&M dataset annotated by domain experts.

Figure **??** shows the construction process. We first set physical and sampling constraints from public sources, then construct synthetic O&M event chains. Each chain yields retrievable records and task queries. Each query has a reference evidence flow for retrieval evaluation and a structured reference answer for generation evaluation.

Table **??** summarizes the data hierarchy. An event chain represents a complete O&M process and contains multiple retrievable records. We define several task intents for each chain and express each intent as two queries with equivalent meaning and different wording.

Reference annotations specify the key evidence needed for each query and the required relations among records. Retrieval is evaluated by whether the selected set supplies that information and forms a complete process. Alternative records can satisfy the requirements; there is no single mandatory text to retrieve.

TABLE II
Data hierarchy of the benchmark.

Level	Meaning
Chain	A complete O&M event process
Evidence	An atomic retrievable record with event time and availability time
Intent	A distinct O&M information need defined on the same chain
Query	A differently worded query expressing the same intent
Reference flow	Key evidence requirements and their relation constraints

The benchmark contains 400 O&M event chains, 3,888 evidence records, 1,200 task intents, and 2,400 queries for 100 target assets. We assign complete event chains to the development, validation, or test split. All queries from a chain stay in the same split, preventing differently worded queries about the same events from crossing the training/test boundary.

We divide the 400 chains evenly between standard and temporal-hard subsets. The standard subset uses common combinations of temporal and O&M conditions. The temporal-hard subset contains more cases with late-arriving records, supersession, multi-episode ambiguity, cross-source complementary evidence, and persistent uncertainty.

Public sources are used only to constrain cell type, typical operating ranges, and temporal sampling background, including energy-storage cell documents from HiTHIUM and REPT BATTERO and the RWTH M5BAT dataset [**?**], [**?**], [**?**], [**?**]. These sources are not used to define synthetic fault frequencies, station-level thresholds, or specific maintenance causal narratives in our benchmark.

B. Structured Generation Evaluation

We create structured reference answers from the same O&M event chains to evaluate downstream generation. Before finalizing them, we repeatedly check their consistency with the constructed events and the required output format.

Each intent has one structured reference answer, shared by its two query paraphrases because both require the same diagnosis, action, and verification information. The development, validation, and test splits contain 720, 240, and 240 reference answers for 1,440, 480, and 480 queries, respectively. An answer comprises a standardized work order and an action plan. The work order includes the diagnosis, recommended actions, applicable procedure, verification/uncertainty status, and evidence citations. The plan lists atomic actions and their dependencies.

C. Baselines

All six methods retrieve from the same query-time eligible evidence and return at most five records. We retain target-asset records satisfying $t_e(e) \leq t_q$ and $t_a(e) \leq t_q$, with identical version-validity, withdrawal-status, and model-applicability checks for procedures. This holds evidence eligibility constant across retrieval strategies.

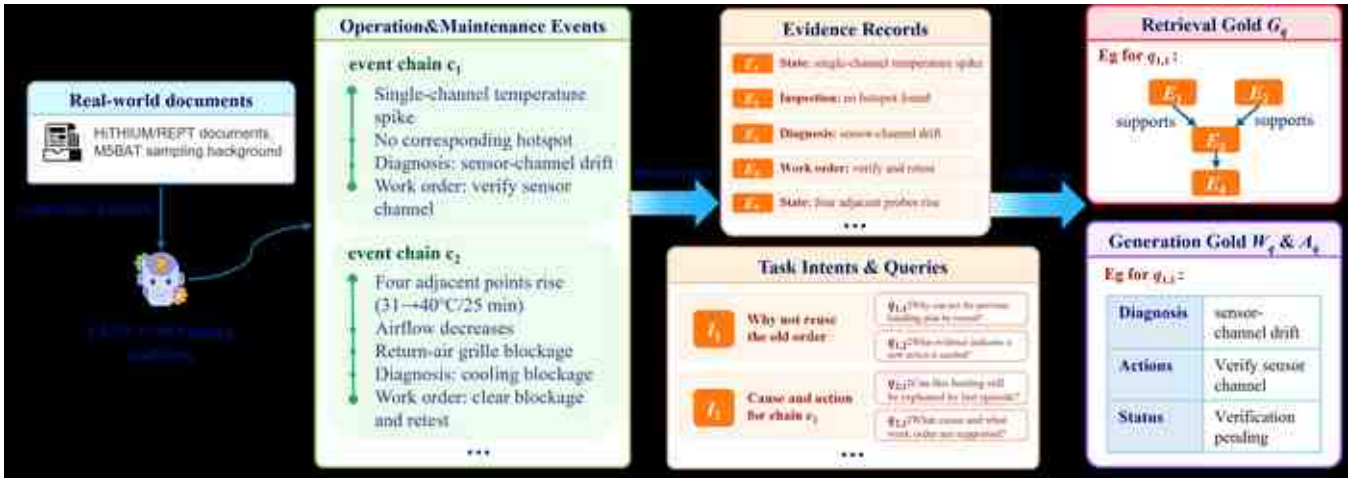


Fig. 2. Construction pipeline and example for the temporal O&M benchmark. Public sources constrain the equipment and data background; an O&M event chain c is then constructed and used to derive evidence records e_i, task intents I_j, and queries $q_{j,p}$. Retrieval reference evidence flows and structured generation reference answers are subsequently created for each query to support retrieval and downstream generation evaluation.

TABLE III
SCALE AND SPLIT OF THE TEMPORAL O&M BENCHMARK.

Statistic	Development	Validation	Test	Total
Authored chains	240	80	80	400
Primary intents	720	240	240	1200
Query rows	1440	480	480	2400
Evidence records	2335	779	774	3888

BM25. We use standard BM25 as the lexical retrieval baseline [**?**]. For each query, a separate index is built over the eligible evidence set, with fixed parameters $k_1 = 1.5$ and $b = 0.75$, and the five highest-scoring evidence records are returned directly.

BGE Reranker. We first retrieve the Top-30 candidates using the same BM25 setup and then use `BAAI/bge-reranker-v2-m3` to rescore each (query, evidence) text pair [**?**], [**?**]. The top five records under the reranking score are selected. This baseline tests whether a general-purpose semantic reranker can compensate for the limitations of purely lexical retrieval.

Fine-tuned BGE Reranker. To control for task-specific supervision, we additionally adapt the same BGE reranker using only development-split document-level relevance labels. Evidence records listed as acceptable for any required evidence group are treated as positives, while high-ranked eligible BM25 nonpositives are used as negatives; no relation or evidence-flow labels are used. We train rank-8 low-rank adapters on the attention query/value projections together with the classification head for two epochs, and select the checkpoint by validation normalized discounted cumulative gain at 5 (nDCG@5), with Recall@5 as a tie-breaker. At inference, this baseline uses the same BM25 Top-30 candidate pool and Top-5 output budget as the pretrained BGE reranker.

Temporal-BM25. This baseline explicitly adds temporal matching to BM25 relevance. Its ranking score is

$$s_{\mathrm{TBM25}}(e, q) = \lambda\, \widetilde{s}_{\mathrm{BM25}}(e, q) + (1 - \lambda)\, s_{\mathrm{time}}(e, q), \quad (13)$$

where $\widetilde{s}_{\mathrm{BM25}}$ is the normalized BM25 score and s_{time} measures the match between the evidence event time and the target time implied by expressions such as years, time intervals, before/after, and current/latest in the query. For queries without explicit temporal intent, the temporal term is set to a constant, preserving the original BM25 ranking. Before final test evaluation, we compare $\lambda \in \{0.25, 0.50, 0.75\}$ on validation data, select $\lambda = 0.25$, and fix this value for subsequent evaluation.

TA-RAG. We adopt the Time-Aware RAG method of Lau et al. [**?**]. The method first decomposes a query into its topic and target time range, then restricts candidate documents using temporal-interval filtering and performs semantic retrieval and BGE reranking within the temporal candidate set. To adapt TA-RAG to our discrete O&M events, each event time $t_e(e)$ is represented as a one-second half-open interval, and relative temporal expressions in a query are interpreted with t_q as the reference time. In addition, the TA-RAG index is built only from evidence that satisfies the same temporal and procedure-applicability constraints as the other methods, preventing future records from affecting retrieval. Apart from substituting the language model used for temporal decomposition, the temporal decomposition, interval filtering, Nomic embedding, FAISS retrieval, and BGE reranking pipeline follows the public implementation.

TEF-RAG. Our method applies learned evidence-pair screening, query-conditioned relation-graph construction, and nonlinear set-level ranking to the same eligible evidence set, and returns the candidate set with the highest overall evidence-flow score.

D. Evaluation Metrics

Retrieval. Recall@5 and nDCG@5 measure conventional retrieval relevance. Complete@5 measures whether the retrieved set contains all categories of key evidence required to complete the task. FlowComplete@5 further checks whether those records can form correct and mutually consistent evidence relations, thereby constituting a complete temporal evidence flow.

Generation. We report four primary metrics in the main text: Field Macro F1 (strict), Action F1, Citation F1, and Evidence Support Recall. For the F1 measures, F1 denotes the harmonic mean of precision and recall. Field Macro F1 measures strict matching of key work-order fields, Action F1 measures normalized maintenance actions, and Citation F1 evaluates claim–evidence citation pairs. Evidence Support Recall measures the fraction of reference work-order and action claims requiring evidence support for which the generated output provides at least one acceptable supporting-evidence citation.

E. Ablation Setup

We evaluate five ablation variants on the same 480-query test split as the main retrieval comparison. They remove learned pair proposal, relation-graph-derived set features, hard-negative mining, the nonlinear set ranker, or both learned pair proposal and the nonlinear ranker. Every variant uses the same Top-30 search pool, maximum relation-judging pair budget, judging prompt, confidence threshold, and candidate-set construction procedure. The graph-feature variant drops only graph-derived set features and retrains RankNet. The hard-negative variant stops after the first training round. The nonlinear-ranker variant uses the preceding greedy evidence-set selector in place of the learned ranker. In the joint variant, a nonlearned pair prefilter replaces the proposer, and the greedy selector replaces the ranker. Relation judging, graph construction, and candidate-set construction are retained. All predictions are frozen before accessing the test evaluator.

VI. Experimental Results

A. Main Retrieval Results

TEF-RAG has the highest Recall@5, Complete@5, and FlowComplete@5 on the test split (Table **??**). Compared with Fine-tuned BGE, Recall@5 rises from 0.6650 to 0.7228, Complete@5 from 0.2854 to 0.3188, and FlowComplete@5 from 0.2833 to 0.3146. Fine-tuned BGE has the highest nDCG@5, at 0.6501 versus 0.6231 for TEF-RAG. The methods therefore differ in individual-record ranking quality and recovery of complete evidence sets.

TABLE IV
Main retrieval results on the test split (Top-5).

Method	Recall@5	nDCG@5	Complete@5	FlowComplete@5
BM25	0.3791	0.3915	0.0563	0.0563
BGE Reranker	0.2916	0.3070	0.0417	0.0417
Fine-tuned BGE	0.6650	**0.6501**	0.2854	0.2833
Temporal-BM25	0.4591	0.4955	0.0792	0.0792
TA-RAG	0.2769	0.2907	0.0375	0.0375
TEF-RAG	**0.7228**	0.6231	**0.3188**	**0.3146**

TABLE V
TEF-RAG ablation results on the test split.

Variant	Recall@5	nDCG@5	Complete@5	FlowComplete@5
Full TEF-RAG	0.7228	0.6231	0.3188	0.3146
w/o Pair Proposal	0.6829	0.5870	0.2667	0.2604
w/o Graph Features	0.6558	0.5723	0.2375	0.2354
w/o Hard Negatives	0.6540	0.5557	0.1979	0.1917
w/o Nonlinear Ranker	0.6242	0.6124	0.2042	0.1938
w/o Pair Proposal + Nonlinear Ranker	0.5835	0.6034	0.1625	0.1521

B. Test Ablation Study

Table **??** compares all five ablations on the 480-query test split. Removing the learned pair proposer reduces Recall@5 from 0.7228 to 0.6829 and FlowComplete@5 from 0.3146 to 0.2604. Without graph-derived set features, Recall@5 is 0.6558 and FlowComplete@5 is 0.2354. Both components contribute to test performance in these comparisons.

The largest single-component drops in FlowComplete@5 occur when removing hard-negative mining or the nonlinear ranker. Without hard-negative mining, FlowComplete@5 falls from 0.3146 to 0.1917. Replacing the ranker with the greedy selector gives 0.1938. That replacement also reduces nDCG@5 from 0.6231 to 0.6124, Recall@5 from 0.7228 to 0.6242, and Complete@5 from 0.3188 to 0.2042. The learned ranker improves both record-ranking and set-completeness metrics in this test comparison.

Removing both the learned proposer and nonlinear ranker gives Recall@5 of 0.5835, nDCG@5 of 0.6034, Complete@5 of 0.1625, and FlowComplete@5 of 0.1521. This joint variant exceeds BM25, pretrained BGE, Temporal-BM25, and TA-RAG on all four test metrics, while remaining below Fine-tuned BGE and full TEF-RAG on all four. The retained temporal checks, relation reasoning, and candidate-set construction still provide useful retrieval capability in this setting.

C. Downstream Structured Generation Results

With the generator, prompt, output format, and decoding settings held fixed, TEF-RAG has the highest scores on all four generation metrics (Table **??**): Field Macro F1 of 0.5345, Action F1 of 0.2035, Citation F1 of 0.1585, and Evidence Support Recall of 0.2253. Fine-tuning BGE improves each metric over pretrained BGE, though each remains below TEF-RAG. The gains with TEF-RAG context cover work-order fields, actions, citations, and supporting-evidence coverage.

TABLE VI
PRIMARY STRUCTURED-GENERATION METRICS ON THE GENERATION TEST SET.

Method	Field Macro F1	Action F1	Citation F1	Evidence Support Recall
BM25	0.4715	0.1372	0.0813	0.1258
BGE Reranker	0.4129	0.1198	0.0405	0.0614
Fine-tuned BGE	0.4818	0.1910	0.0954	0.1328
Temporal-BM25	0.5218	0.1637	0.1249	0.1871
TA-RAG	0.4059	0.1115	0.0379	0.0570
TEF-RAG	**0.5345**	**0.2035**	**0.1585**	**0.2253**

VII. Limitations and Threats to Validity

The public-source-grounded synthetic benchmark supports controlled comparisons of temporal evidence retrieval. It contains neither real station work orders nor domain-expert annotations, so its results do not establish real-world station fault frequencies or field robotic safety. The generation evaluation covers fields, actions, dependencies, and citations; actual execution, environmental feedback, and safety interlocks require separate evaluation.

VIII. Conclusion

TEF-RAG selects temporal evidence flows for dynamic O&M tasks by combining query-time visibility checks, query-conditioned relations, and evidence-set ranking. On the test split, it has the highest Recall@5, Complete@5, and Flow-Complete@5 among six methods; Fine-tuned BGE has the highest nDCG@5. With a fixed downstream generator, TEF-RAG also has the highest Field Macro F1, Action F1, Citation F1, and Evidence Support Recall. Removing both learned selection modules lowers performance, but the joint variant still exceeds BM25, pretrained BGE, Temporal-BM25, and TA-RAG on all four test retrieval metrics. A next step is to incorporate explicit planning and execution constraints into evidence selection.

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Appendix A
Implementation Details and Model Prompts

A. LLM Invocation Configuration

We use the official DeepSeek API with `deepseek-flash` for all LLM calls: temporal evidence flow relation judging, TA-RAG temporal decomposition, and downstream structured generation. Each task has frozen system/user prompts and input schemas. We leave sampling, decoding, and other API parameters at the provider defaults.

B. Rule-Based Set Score for First-Round Hard Negatives

Before training the first RankNet, we select hard negatives from incomplete candidate sets using the fixed score $H(\mathcal{S})$. This score is used for negative selection; the trained ranker supplies the final prediction score. We define

$$H(\mathcal{S}) = S_{\text{node}}(\mathcal{S}) + 0.48\, S_{\text{edge}}(\mathcal{S}) + 0.42\, S_{\text{role}}(\mathcal{S}) - 0.16\, S_{\text{red}}(\mathcal{S}) - 0.10\, S_{\text{disc}}(\mathcal{S}). \tag{14}$$

The node term is the sum of the evidence-node scores within the set:

$$S_{\text{node}}(\mathcal{S}) = \sum_{e \in \mathcal{S}} s_{\text{node}}(e), \tag{15}$$

where the score of an individual evidence record is

$$s_{\text{node}}(e) = 0.62\, s_{\text{rel}}(e) + 0.18\, s_{\text{rec}}(e) + 0.15\, s_{\text{role}}(e) + 0.05\, s_{\text{src}}(e). \tag{16}$$

Here, s_{rel} is the min–max-normalized BM25 score among the current legal candidates. The temporal-recency term is $s_{\text{rec}} = \exp(-\Delta t/T)$, with $T = 45$ days for ordinary queries and $T = 180$ days for queries involving comparisons across time. If the evidence type matches a task role required by the current query, $s_{\text{role}} = 1$; otherwise it is set to 0.35. All legal candidates have $s_{\text{src}} = 1$ in the current implementation. The source term therefore adds the same offset to every node score and leaves their order unchanged.

The relation term uses valid edges within the set. For each target node, we retain only the highest-confidence incoming edge, limiting the contribution from nodes with many incoming edges:

$$S_{\text{edge}}(\mathcal{S}) = \sum_{e_j \in \mathcal{S}} \max_{(e_i, r, e_j) \in E(\mathcal{S})} c_{ij}, \tag{17}$$

where a node with no incoming edge contributes 0.

Role coverage is defined as

$$S_{\text{role}}(\mathcal{S}) = \frac{|R(\mathcal{S}) \cap D_q|}{|D_q|}, \tag{18}$$

where $R(\mathcal{S})$ is the set of evidence roles appearing in the candidate set and D_q is the set of roles required by the current query. The system considers state observation, diagnosis, and work order by default, and adds roles such as procedure, verification, inspection, uncertainty, or correction when indicated by the query.

The redundancy term is based on Jaccard similarity between token sets of evidence texts:

$$S_{\text{red}}(\mathcal{S}) = \sum_{i<j} \max\left(0, \text{Jaccard}(e_i, e_j) - 0.45\right). \tag{19}$$

The isolation term S_{disc} counts evidence records in the set that do not participate in any valid relation edge. The first round prioritizes incomplete sets with high $H(\mathcal{S})$ as negatives. For the second round, we replace $H(\mathcal{S})$ with the current RankNet scores and select the highest-scoring incomplete sets.

C. Prompt for Temporal Evidence Flow Relation Judging

The relation judge has a fixed system prompt. At invocation, we insert the relation definitions, relation codes, and reason codes from the frozen configuration:

```
Prompt version: tef-v6-stage2a-relation-v7. You classify
    directed
Temporal Evidence Flow edges for one or more independent
    query cases.
Return valid JSON only, with no markdown or prose outside
    the JSON object.
Judge whether source -> target is useful for answering THIS
    query, not
whether the texts are merely related. Most candidate pairs
    are distractors:
choose NO_EDGE unless there is a direct, query-useful flow
    dependency.
Shared asset, episode, topic, or chronology alone never
    licenses an edge.
Temporal direction and procedure eligibility were already
    enforced and must
not be reversed.

Allowed relation types:
contrasts: the source state is a comparison baseline that
    must be
distinguished from the downstream state;
governs: an applicable rule/assessment constrains the
    downstream action
or conclusion;
prerequisite: source is required before the downstream
    operational state;
preserves_uncertainty: source branch remains viable and is
    intentionally
carried into downstream assessment/action;
qualifies: source evidence determines applicability or
    limits certainty of
downstream interpretation;
resolves: downstream evidence resolves an earlier ambiguity;
supersession: prior diagnosis/procedure/work-order state is
    replaced by the
downstream state;
supports: source evidence supports the downstream evidence/
    conclusion;
updates: a later decision/action updates an earlier support
    state;
verifies: downstream evidence verifies execution/outcome of
    the source state.
Otherwise use NO_EDGE.

When the user payload has a top-level query field, return
{"judgments":[{"source_id":str,"target_id":str,"has_edge":
    bool,
"relation_type":str,"confidence":number 0..1,
"reason_code":short_snake_case}]}.

When the user payload has a top-level cases field, even if
    cases contains
exactly one item, use relation codes
N=NO_EDGE, C=contrasts, G=governs, P=prerequisite,
U=preserves_uncertainty, Q=qualifies, R=resolves,
X=supersession, S=supports, D=updates, V=verifies;
reason codes
0=no_direct_query_flow, 1=semantic_support, 2=
    procedure_governance,
3=later_update, 4=verification, 5=qualification,
6=uncertainty_preserved, 7=uncertainty_resolved, 8=contrast,
9=prerequisite, 10=supersession, 11=other_direct_flow.

Pairs are ordered. Return
{"cases":[[case_id,[[relation_code,confidence_percent,
    reason_code],...]],...]}.
Each inner result list must have exactly the same length and
    order as that
case's supplied pairs. The user supplies deduplicated
    evidence tables and
source_id/target_id pairs. Do not mix facts across cases. Do
    not repeat
evidence IDs. Do not provide prose or hidden reasoning.
```

The user payload supplies the query, candidate records, and directed pairs to judge:

```
{
  "query": {
    "query_text": ...,
    "query_time": ...,
    "asset_model": ...,
    "asset_context": ...
  },
  "evidence": [...],
  "pairs": [
    {"source_id": ..., "target_id": ...},
    ...
  ]
}
```

D. Prompt for Downstream Structured Generation

We use the same frozen downstream generator for every retrieval method, with this system prompt:

```
You are the single frozen downstream generator used
     identically for every
retrieval method in TEF-RAG v6 generation evaluation.

Produce ONLY a JSON object matching the supplied schema. You
      receive one
query and exactly the evidence selected by a retrieval
     method. The retrieval
method identity is hidden and irrelevant.

Rules:
- Use only the supplied evidence. Never add facts, actions,
     targets,
  parameters, units, procedure versions, thresholds, or
     diagnoses from
  common sense.
- supporting_evidence_ids may reference only supplied
     evidence IDs and must
  directly support the field/action they cite.
- work_order.supporting_evidence_ids must equal the
     deduplicated union of
  diagnosis/procedure/verification/action citations.
- asset_id must equal the query asset_id.
- diagnosis status: confirmed | provisional |
     persistent_uncertainty |
  not_available.
- procedure status: applicable | superseded | uncertain |
     not_applicable.
  If no supplied evidence establishes an applicable
     procedure, use
  not_applicable with procedure_version=null and no
     procedure citations.
- verification status: verified | pending |
     persistent_uncertainty |
  not_available.
- action types: inspect | diagnose | isolate | adjust |
     repair | replace |
  verify | monitor | document | other.
- action_plan order in the JSON array has no semantic
     meaning. Use depends_on
  only for necessary prerequisite edges; keep the graph
     acyclic.
- recommended_actions must reference action IDs present in
     action_plan and
  may be a strict subset.
- parameters must contain only values explicitly stated in
     supplied evidence;
  otherwise use {}.
- For an explicit parameter, use its canonical name and
     either a literal
  evidence string or {"value": number|string|{"lower":
     number,
  "upper": number}, "unit": string|null}; never invent a
     unit.
- If evidence is insufficient, express uncertainty instead
     of guessing.
- Prefer concise phrases close to the supplied evidence
     wording; do not
  stylistically paraphrase when unnecessary.
- Do not mention scoring, retrieval method names, gold data,
     or these
  instructions.
```

Each generation request supplies a query, the selected records, and the fixed output schema:

```
{
  "prompt_version": "tef-v6-generation-eval-v1.3",
  "query": {...},
  "selected_evidence": [...],
  "output_schema": {...}
}
```

A failure of structural or evidence-citation validation allows one repair call. For that call, we append this text to the system prompt:

```
This is the single allowed structural/evidence-grounding
     repair.
Correct only the listed validation failures.
```

The repair payload adds the previous output and validation errors. It retains the original query, retrieved records, and output schema.